

C Programming Mini Project Report

1. Problem Statement

Section	Description
Existing Problem	Cinema ticket counters often allocate seats manually or using printed charts, making it difficult to track real-time seat availability, avoid double-booking, and calculate revenue accurately.
Proposed Solution	Develop a console-based Cinema Seat Reservation System using C Programming to automate seat booking, cancellation, searching, and revenue tracking.
Target Users	Cinema Booking Counter Staff, Theatre Managers
Expected Benefits	Faster seat booking, elimination of double-booking, quick lookup of customer bookings, and instant revenue/occupancy summaries.

2. Project Objectives

Objective ID	Objective
OBJ-01	Display the current seat availability map
OBJ-02	Book a seat for a customer
OBJ-03	Cancel an existing booking
OBJ-04	Search bookings by customer name
OBJ-05	Generate a booking summary (seats booked/available, revenue)
OBJ-06	Validate seat input and prevent invalid or duplicate bookings
OBJ-07	Apply differential pricing for Premium and Regular rows

3. User Stories

ID	User Story	Acceptance Criteria
US-01	As a staff member, I want to view the seat map so I can see which seats are booked and which are free.	Booked seats are shown blank; free seats display their seat code (e.g., A1).
US-02	As a staff member, I want to book a seat for a customer so the booking is recorded.	Seat is marked booked and the customer name is stored; already-booked seats are rejected.
US-03	As a staff member, I want to cancel a booking so the seat becomes available again.	Seat is freed and the customer name is cleared.
US-04	As a staff member, I want to search bookings by customer name so I can quickly find a customer's seat.	All seats booked under the given name are displayed; a clear message is shown if none are found.
US-05	As a manager, I want to view a summary of bookings so I can track occupancy and revenue.	Total seats, booked seats, available seats, and total revenue are calculated correctly.

4. Functional Requirements

Requirement ID	Requirement	Status
FR-01	View Seat Availability	✓
FR-02	Book a Seat	✓
FR-03	Cancel a Booking	✓
FR-04	Search Booking by Customer Name	✓
FR-05	View Booking Summary (Seats & Revenue)	✓
FR-06	Seat Input Validation	✓
FR-07	Differential Pricing (Premium / Regular)	✓

5. Project Modules

Module	Description	C Concepts Used
Seat Map Display	Show the current booking status of all seats	Arrays, Loops, Functions
Seat Booking	Book a seat and record the customer name	Arrays, Strings, Functions
Seat Cancellation	Cancel an existing booking	Arrays, Functions
Booking Search	Search bookings by customer name	Loops, String Comparison
Booking Summary	Calculate seats booked/available and revenue	Loops, Arithmetic
Menu System	Handle user navigation between options	Switch Case, Loops

6. C Programming Concepts Used

Concept	Implementation
Variables	Store row/seat indices, prices, and menu choices
Data Types	int, char
Arrays	2D array for booking status; 3D array for customer names per seat
Functions	Modular design (displaySeatMap, bookSeat, cancelSeat, searchBooking, showSummary, etc.)
Loops	Menu-driven operation and iteration over the seat grid
Switch Case	Menu selection
Conditional Statements	Seat validation and booking status checks
String Handling	Customer name storage and comparison (string.h, ctype.h)

7. Expected Program Features

Feature	Description
View Seat Availability	Display the seat map with booked/free status and pricing per row
Book a Seat	Reserve a seat for a customer by row letter and seat number
Cancel a Booking	Free up a previously booked seat
Search Booking	Find all seats booked under a given customer name
View Booking Summary	Show total seats, seats booked, seats available, and revenue
Exit	Close the application safely